

Basic Matrix Algebra for Statistical Modeling

This chapter is unique in that no new statistical procedures are introduced. Rather, the current chapter serves to introduce matrix algebra in the gentlest way possible. Following a discussion why it is useful to know a little about matrix algebra, the main topics of the chapter include:

- Kinds of matrices and simple matrix operations
- Matrix addition and subtraction
- Multiplication with matrices
- Determinants and matrix inversion
- Matrix calculations for statistical applications
- Matrix equations for linear regression
- Eigenvalues and eigenvectors

Why matrix algebra?

Chapters 1 to 7 of the text focused on models for a single outcome variable Y . But moving forward, most of the models presented in Chapters 8 to 12 are truly **multivariate** in nature in that there will be more than one dependent or outcome variable; that is, we will have $Y_1, Y_2, \dots, Y_Q$ outcomes instead of a single Y . As a result, there will be a separate equation for each of the Q outcomes, but the parameters in the equations will all be estimated *simultaneously* rather than through one-at-a-time, separate analyses of each Y .

When the number of outcome variables Q for a given multivariate model becomes large, it can be onerous to write out the entire model using a separate equation for each Y . Instead, being able to express mathematical relationships using *matrices* greatly facilitates our ability to describe multivariate statistical methods; a set of Q separate equations can often be expressed using a single, compact matrix equation. Additionally, univariate models for only a single outcome variable can be understood as a special case of a multivariate model, and therefore also are represented within a more general matrix framework. Matrix operations form the foundation for all statistical computing. The statistical procedures implemented in software such as R, SAS, and SPSS, even the simple ones, are all programmed in terms of matrix operations. To demonstrate, this chapter will present the matrix formulas for simple descriptive statistics and OLS regression.

This chapter provides only a light introduction to the standard notational conventions and algebraic operations for matrix algebra as they pertain to statistical modeling. Having these conventions under our belt will allow presentation of modeling procedures in Chapters 8 to 12 in a concise fashion which is consistent with how these methods are represented in the behavioural and social science methodological literature as well as within certain computer software implementations. My hope is that this material will also give readers a certain level of comfort with matrix formulas that will enhance their comprehension of methodological research papers, including those I cite throughout this text. At its core, the mathematical background needed for this chapter is minimal and really involves nothing more complex than the simple algebra of the previous chapters; there is still no calculus involved. The main foci of this chapter are definitions, notation, and mathematical rules for matrices. Some details and examples of matrix computations are presented in this chapter, but given their complexity and tedium, the work of matrix calculation is best left for computers. Readers who desire more comprehensive treatments of matrix algebra and its application to statistics are referred to Fox (2009) and Carroll et al. (1997).

Kinds of Matrices and Simple Matrix Operations

A **matrix** is an organized, rectangular array of quantities known as **elements**. Most often, an element is a single **scalar** term, such as a numerical value or an algebraic symbol representing some numerical value¹. For an initial example matrix, consider matrix **A**:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} 1.0 & 3.0 \\ 0.2 & 4.6 \\ 8.5 & -2.3 \end{bmatrix}.$$

It is a standard notational convention to symbolize a matrix with a letter in bold-faced font (e.g., **A**), whereas an individual scalar term is symbolized in lower-case, italic font (e.g., a_{11}). Element a_{ij} of **A** is the entry in the i th row and j th column. Thus, $a_{11} = 1.0$ is the entry in the first row, first column of **A**, $a_{12} = 3.0$ is the entry in the first row, second column, $a_{21} = 0.2$ is in the second row, first column, and so on. A matrix is characterized by the number of rows, I , and

¹ It is possible, though, to have a matrix of matrices, which is known as a *block matrix*. An element of a block matrix is another smaller matrix, that is, a subset of the larger block matrix.

the number of columns, J , which define the matrix's **order** or **dimension**. Above, matrix **A** has dimension (3×2) because there are $I = 3$ rows and $J = 2$ columns².

If a research study has N cases (e.g., N research participants) and P variables, then the data for the study are typically saved in a **data matrix** of dimension $N \times P$. That is, when you open a data file in a program such as SPSS or Excel, you are looking at an $N \times P$ data matrix, although you may not explicitly think of it as a matrix. Other examples of matrices with which you are likely already familiar are correlation matrices and covariance matrices, which are described in detail later in this chapter.

The **transpose** of a given matrix **X**, denoted **X'** (or **X^T**), is formed by writing the rows of **X** as the columns of **X'**. Thus, the transpose of matrix **A** given above is

$$\mathbf{A}' = \begin{bmatrix} 1.0 & 0.2 & 8.5 \\ 3.0 & 4.6 & -2.3 \end{bmatrix}$$

For any matrix, $(\mathbf{X}')' = \mathbf{X}$. The ability to transpose a matrix is particularly useful for making matrices *conformable* for multiplication, which is described later.

A matrix consisting of only one column is called a **column vector**. Vectors are often symbolized with a bold letter in lower-case. For example,

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_I \end{bmatrix}$$

is a generic $(I \times 1)$ column vector. Similarly, a matrix consisting of only one row is called a **row vector**, such as the $(1 \times J)$ vector **b'**:

$$\mathbf{b}' = [b_1 \quad b_2 \quad \cdots \quad b_J] .$$

When a vector is a row vector, it is common to attach the “prime” symbol to the vector name (which is why the row vector above is called **b'** instead of just **b**), which indicates that the row vector is the transpose of some column vector. By convention, a vector is generally assumed to be a column vector (such as **a** above) unless it is explicitly written as a transpose (such as **b'**).

A **square matrix** of dimension J , as the name implies, has J rows and J columns. The entries c_{jj} (that is, c_{B11B} , c_{B22B} , ..., c_{BJJB}) of a square matrix **C** comprise the main **diagonal** of the

² A single scalar value can be considered a (1×1) matrix.

matrix. Thus, by definition, the diagonal goes from the top-left of a square matrix to the bottom-right. For example,

$$\mathbf{C} = \begin{bmatrix} -5 & 1 & 3 \\ 2 & 2 & 6 \\ 7 & 3 & -4 \end{bmatrix}$$

is a square matrix of dimension $J = 3$ because the number of rows (3) equals the number of columns (3). The elements of the diagonal are $c_{11} = -5$, $c_{22} = 2$, and $c_{33} = -4$. The elements above the diagonal of a square matrix (here, $c_{12} = 1$, $c_{13} = 3$, and $c_{23} = 6$) form the **upper triangle** while the elements below the diagonal (here, $c_{21} = 2$, $c_{31} = 7$, and $c_{32} = 3$) form the **lower triangle**. The **trace** of a square matrix $\mathbf{C}$, denoted $\text{tr}(\mathbf{C})$, equals the sum of its diagonal elements. Given the current $\mathbf{C}$, we have

$$\text{tr}(\mathbf{C}) = \sum_{j=1}^J c_{jj} = -5 + 2 + -4 = -7.$$

Next, a square matrix $\mathbf{A}$ is **symmetric** if $\mathbf{A} = \mathbf{A}'$, that is, if $a_{ij} = a_{ji}$ for all i and j .

Consequently, the matrix $\mathbf{C}$ above is *not* symmetric (e.g., because $c_{12} \neq c_{21}$, $c_{13} \neq c_{31}$, etc.) But the matrix $\mathbf{D}$,

$$\mathbf{D} = \begin{bmatrix} -5 & 1 & 3 \\ 1 & 2 & 6 \\ 3 & 6 & -4 \end{bmatrix},$$

is symmetric because $\mathbf{D} = \mathbf{D}'$ (the elements in the upper triangle are a mirror image of the elements in the lower triangle; $d_{12} = d_{21}$, $d_{13} = d_{31}$, and $d_{23} = d_{32}$).

All **correlation matrices**, which are usually denoted $\mathbf{R}$, are square, symmetric matrices with each diagonal element equal to 1. For example, the correlations given in Chapter 2 in Table 2.1 can be organized in a matrix as

$$\mathbf{R} = \begin{bmatrix} 1.00 & 0.32 & -0.21 \\ 0.32 & 1.00 & -0.14 \\ -0.21 & -0.14 & 1.00 \end{bmatrix}.$$

But because a correlation matrix is symmetric, usually only the lower triangle is displayed when the correlation matrix is presented as a table in a research report, as in the original Table 2.1:

	<i>profit</i>	<i>review</i>	<i>length</i>
<i>profit</i>	1		
<i>review</i>	.30	1	
<i>length</i>	.18	.30	1

Or, equivalently, one might present only the upper triangle of a symmetric matrix:

	<i>profit</i>	<i>review</i>	<i>length</i>
<i>profit</i>	1	.30	.18
<i>review</i>		1	.30
<i>length</i>			1

Because each diagonal element of a correlation matrix equals 1, the trace of a correlation matrix equals the total number of variables in the correlation matrix; in the current example, $\text{tr}(\mathbf{R}) = 3$.

Recall from Equation 1.9 that a correlation is simply a covariance that has been standardized according to the standard deviations of the two variables being correlated:

$$r_{XY} = \frac{\text{COV}(X, Y)}{\sqrt{\text{VAR}(X)}\sqrt{\text{VAR}(Y)}} = \frac{s_{XY}}{s_X s_Y}.$$

Therefore, the covariances among a set of two or more variables can also be contained within a square, symmetric matrix, a **covariance matrix**, which is often denoted $\mathbf{S}$. For example, with three variables Y , X_1 , and X_2 , the (population) covariance matrix is

$$\mathbf{S} = \begin{bmatrix} \text{VAR}(Y) & \text{COV}(Y, X_1) & \text{COV}(Y, X_2) \\ \text{COV}(Y, X_1) & \text{VAR}(X_1) & \text{COV}(X_1, X_2) \\ \text{COV}(Y, X_2) & \text{COV}(X_1, X_2) & \text{VAR}(X_2) \end{bmatrix} = \begin{bmatrix} \sigma_Y^2 & \sigma_{YX_1} & \sigma_{YX_2} \\ \sigma_{YX_1} & \sigma_{X_1}^2 & \sigma_{X_1X_2} \\ \sigma_{YX_2} & \sigma_{X_1X_2} & \sigma_{X_2}^2 \end{bmatrix}.$$

Whereas the diagonal elements of a *correlation* matrix are all equal to 1, the diagonal elements of a *covariance* matrix are the variances of the variables.

Moving on, a square matrix is called a **diagonal matrix** if all elements equal 0 except those on the main diagonal:

$$\mathbf{D} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_j \end{bmatrix}.$$

An especially important type of diagonal matrix is an **identity** matrix $\mathbf{I}$, in which each diagonal element equals 1:

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

Another important type of matrix (which need not be square) is a **zero matrix**, $\mathbf{0}$, in which *all* elements equal zero:

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Elementary Matrix Algebra

Matrix addition and subtraction

The major strength, indeed the very purpose, of matrix algebra is that it provides a framework for algebraically manipulating large amounts of numerical entities using small, concise matrix expressions. As we will see, a set of many equations expressed using scalar terms can usually be re-expressed using only a few (or even only one) matrix equation. The cost of this efficiency is that matrix algebra adheres to a strict set of rules, some of which are immediately intuitive, and some of which are not.

Two matrices can be added *only* if they have the same dimension; then their sum is formed by adding corresponding elements. Thus, if $\mathbf{A}$ and $\mathbf{B}$ are both of dimension $(I \times J)$, then their sum, $\mathbf{C} = \mathbf{A} + \mathbf{B}$, is also of dimension $(I \times J)$, with $c_{ij} = a_{ij} + b_{ij}$. For example, matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

and

$$\mathbf{B} = \begin{bmatrix} -5 & 1 & 2 \\ 3 & 0 & -4 \end{bmatrix}$$

both have dimension (2×3) , and so their sum is another (2×3) matrix

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \begin{bmatrix} 1+(-5) & 2+1 & 3+2 \\ 4+3 & 5+0 & 6+(-4) \end{bmatrix} = \begin{bmatrix} -4 & 3 & 5 \\ 7 & 5 & 2 \end{bmatrix}.$$

Likewise, if $\mathbf{D} = \mathbf{A} - \mathbf{B}$, then $\mathbf{D}$ is also of dimension $(I \times J)$, with $d_{ij} = a_{ij} - b_{ij}$:

$$\mathbf{D} = \mathbf{A} - \mathbf{B} = \begin{bmatrix} 1-(-5) & 2-1 & 3-2 \\ 4-3 & 5-0 & 6-(-4) \end{bmatrix} = \begin{bmatrix} 6 & 1 & 1 \\ 1 & 5 & 10 \end{bmatrix}.$$

Furthermore, the negative of a matrix $\mathbf{B}$, such as $\mathbf{E} = -\mathbf{B}$, is of the same dimension as $\mathbf{B}$, with elements $e_{ij} = -b_{ij}$:

$$\mathbf{E} = -\mathbf{B} = \begin{bmatrix} 5 & -1 & -2 \\ -3 & 0 & 4 \end{bmatrix}.$$

Because matrix addition, subtraction, and negation are performed in an *element-wise* way (e.g., for addition, element i,j from one matrix is added to element i,j from the other matrix, and so on, across all $I \times J$ elements), essentially the same intuitive laws are obeyed as for the corresponding scalar operations. Specifically, matrix addition is *commutative* in that

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

and matrix addition is *associative* in that

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}).$$

Additionally, negation of a matrix distributes through parentheses in the same way as with scalar algebra, and so

$$-(-\mathbf{A}) = \mathbf{A}$$

and

$$(\mathbf{A} - \mathbf{B}) = \mathbf{A} + (-\mathbf{B}) = -(\mathbf{B} - \mathbf{A}).$$

It is also intuitively clear that subtracting a matrix from itself produces a zero matrix,

$$\mathbf{A} - \mathbf{A} = \mathbf{0},$$

and that

$$\mathbf{A} + \mathbf{0} = \mathbf{A}.$$

Finally, it is also the case that

$$(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}',$$

that is, the transpose of a sum of two matrices equals the sum of the two matrices individually transposed.

Multiplication with matrices

Like matrix addition and subtraction, the operation and rules for multiplying a scalar term by a matrix are also relatively straightforward. The product of a scalar c and a $(I \times J)$ matrix $\mathbf{A}$ is an $(I \times J)$ matrix $\mathbf{B} = c\mathbf{A}$ in which $b_{ij} = c*a_{ij}$. For example, if $c = 3$ and

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 8 & -2 \end{bmatrix},$$

then their product is

$$\mathbf{B} = 3 * \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 8 & -2 \end{bmatrix} = \begin{bmatrix} 3*1 & 3*3 \\ 3*2 & 3*5 \\ 3*8 & 3*-2 \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 6 & 15 \\ 24 & -6 \end{bmatrix}.$$

The product of a scalar and a matrix is also commutative such that

$$c\mathbf{A} = \mathbf{A}c,$$

and *distributive* such that

$$\mathbf{A}(b + c) = \mathbf{A}b + \mathbf{A}c$$

and

$$c(\mathbf{A} + \mathbf{B}) = c\mathbf{A} + c\mathbf{B}.$$

Finally, it is also clear that

$$0\mathbf{A} = \mathbf{0},$$

$$1\mathbf{A} = \mathbf{A},$$

and

$$(-1)\mathbf{A} = -\mathbf{A}.$$

Whereas these rules for multiplying a matrix by a scalar are straightforward, the operation and rules for multiplying two matrices together are considerably more complex. First, it might not even be possible to multiply two given matrices together; two matrices must be **conformable** for multiplication to be possible. Two matrices $\mathbf{A}$ and $\mathbf{B}$ are *conformable* for multiplication in the order given (i.e., $\mathbf{AB}$ rather than $\mathbf{BA}$) if the number of columns of the left-hand matrix ($\mathbf{A}$) is equal to the number of rows of the right-hand matrix ($\mathbf{B}$). Thus, $\mathbf{A}$ and $\mathbf{B}$ are conformable for multiplication if $\mathbf{A}$ has dimension $(I \times J)$ and $\mathbf{B}$ is of dimension $(J \times K)$. For example, the two matrices below are conformable:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix},$$

meaning that is possible to calculate the product $\mathbf{AB}$ because the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$. But it is *not* possible to calculate the product of these two matrices as $\mathbf{BA}$ because the number of columns of $\mathbf{B}$ does not equal the number of rows of $\mathbf{A}$. It is possible, though, to calculate $\mathbf{BA}'$, which shows the usefulness of transposing a matrix.

Unlike addition and subtraction of matrices and the multiplication of a matrix by a scalar, the operation of multiplying two matrices together is not simply element-wise. Let $\mathbf{C} = \mathbf{AB}$ be

the matrix product of $(I \times J)$ matrix $\mathbf{A}$ and $(J \times K)$ matrix $\mathbf{B}$; also, let $\mathbf{a}'_i$ represent the i th row of $\mathbf{A}$ and $\mathbf{b}_j$ represent the j th column of $\mathbf{B}$. Then $\mathbf{C}$ is a matrix of dimension $(J \times K)$ with elements calculated as

$$c_{ij} = \mathbf{a}'_i \cdot \mathbf{b}_j = \sum_{n=1}^J a_{in} b_{nj}.$$

Element c_{ij} is known as the **dot product** of row i from matrix $\mathbf{A}$ and column j from matrix $\mathbf{B}$. For example, using matrices $\mathbf{A}$ and $\mathbf{B}$ in the previous paragraph, because $\mathbf{A}$ has dimension (3×2) and $\mathbf{B}$ has dimension (2×2) , then their product $\mathbf{C}$ will be (3×2) . Thus the six elements of $\mathbf{C}$ are c_{11} , c_{12} , c_{21} , c_{22} , c_{31} , and c_{32} , each of which is a dot product of the corresponding row of $\mathbf{A}$ and column of $\mathbf{B}$. The dot product of the first row of $\mathbf{A}$ and the first column of $\mathbf{B}$ is

$$c_{11} = \mathbf{a}'_1 \cdot \mathbf{b}_1 = (2 \ 1) \cdot (1 \ 3) = (2 \times 1) + (1 \times 3) = 5.$$

Next, the dot product of the first row of $\mathbf{A}$ and the second column of $\mathbf{B}$ is

$$c_{12} = \mathbf{a}'_1 \cdot \mathbf{b}_2 = (2 \ 1) \cdot (3 \ 1) = (2 \times 3) + (1 \times 1) = 7.$$

The remaining dot products are

$$c_{21} = \mathbf{a}'_2 \cdot \mathbf{b}_1 = (4 \ 2) \cdot (1 \ 3) = (4 \times 1) + (2 \times 3) = 10,$$

$$c_{22} = \mathbf{a}'_2 \cdot \mathbf{b}_2 = (4 \ 2) \cdot (3 \ 1) = (4 \times 3) + (2 \times 1) = 14,$$

$$c_{31} = \mathbf{a}'_3 \cdot \mathbf{b}_1 = (6 \ 3) \cdot (1 \ 3) = (6 \times 1) + (3 \times 3) = 15,$$

and

$$c_{32} = \mathbf{a}'_3 \cdot \mathbf{b}_2 = (6 \ 3) \cdot (3 \ 1) = (6 \times 3) + (3 \times 1) = 21.$$

Therefore, the resulting matrix product is

$$\mathbf{AB} = \mathbf{C} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 10 & 14 \\ 15 & 21 \end{bmatrix}.$$

For another example, if instead we have

$$\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } \mathbf{E} = \begin{bmatrix} 0 & 3 \\ 2 & 1 \end{bmatrix},$$

because $\mathbf{D}$ is (2×2) and $\mathbf{E}$ is (2×2) , then their product $\mathbf{F}$ will also be (2×2) . The dot product of the first row of $\mathbf{D}$ and the first column of $\mathbf{E}$ is

$$f_{11} = \mathbf{d}'_1 \cdot \mathbf{e}_1 = (1 \ 2) \cdot (0 \ 2) = (1 \times 0) + (2 \times 2) = 4.$$

Next, we have

$$f_{12} = \mathbf{d}'_1 \cdot \mathbf{e}_2 = (1 \ 2) \cdot (3 \ 1) = (1 \times 3) + (2 \times 1) = 5,$$

$$f_{21} = \mathbf{d}'_2 \cdot \mathbf{e}_1 = (3 \ 4) \cdot (0 \ 2) = (3 \times 0) + (4 \times 2) = 8,$$

and finally

$$f_{22} = \mathbf{d}'_2 \cdot \mathbf{e}_2 = (3 \ 4) \cdot (3 \ 1) = (3 \times 3) + (4 \times 1) = 13.$$

Therefore

$$\mathbf{F} = \begin{bmatrix} 4 & 5 \\ 8 & 13 \end{bmatrix}.$$

Matrix multiplication is generally *not* commutative. That is, it is often (but not always) the case that $\mathbf{AB} \neq \mathbf{BA}$. For example, using matrices $\mathbf{D}$ and $\mathbf{E}$ above, when we **post-multiplied** $\mathbf{D}$ by $\mathbf{E}$, we obtained

$$\mathbf{F} = \mathbf{DE} = \begin{bmatrix} 4 & 5 \\ 8 & 13 \end{bmatrix}.$$

But if we instead **pre-multiply** $\mathbf{D}$ by $\mathbf{E}$, the result is

$$\mathbf{ED} = \begin{bmatrix} 9 & 12 \\ 5 & 8 \end{bmatrix} \neq \mathbf{F}.$$

Other rules for matrix multiplication follow more intuitively. Matrix multiplication is associative, meaning that $\mathbf{A(BC)} = (\mathbf{AB})\mathbf{C}$. Furthermore, matrix multiplication is also *distributive* with respect to addition in that

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

and

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}.$$

Next, the identity and zero matrices play roles with respect to matrix multiplication analogous to those of the numbers 0 and 1 in scalar algebra. In particular,

$$\underset{(I \times J)(J \times J)}{\mathbf{A}} \underset{(I \times I)(I \times J)}{\mathbf{I}} = \underset{(I \times I)(I \times J)}{\mathbf{I}} \underset{(I \times J)}{\mathbf{A}} = \mathbf{A},$$

meaning that the post- or pre-multiplying any matrix $\mathbf{A}$ by the conformable identity matrix results in the same original matrix $\mathbf{A}$. Furthermore,

$$\underset{(I \times J)(J \times K)}{\mathbf{A}} \underset{(I \times K)}{\mathbf{0}} = \underset{(I \times K)}{\mathbf{0}}$$

and

$$\underset{(M \times I)(I \times J)}{\mathbf{0}} \underset{(M \times J)}{\mathbf{A}} = \underset{(M \times J)}{\mathbf{0}},$$

meaning that the post- or pre-multiplying any matrix $\mathbf{A}$ by the conformable zero matrix results in another zero matrix. A further property of matrix multiplication is that $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$; that is, the transpose of a product is the product of the transposes taken in the opposite order. Finally, the *powers* of a *square* matrix are the products of the matrix with itself. That is,

$$\mathbf{A}^2 = \mathbf{A}\mathbf{A},$$

$$\mathbf{A}^3 = \mathbf{A}\mathbf{A}\mathbf{A} = \mathbf{A}\mathbf{A}^2 = \mathbf{A}^2\mathbf{A}, \text{ and so on.}$$

What about division? Determinants and matrix inversion

For any square matrix $\mathbf{A}$, a scalar, numeric value known as the **determinant** can be calculated, which is symbolized as $|\mathbf{A}|$ or $\det(\mathbf{A})$. Determinants are used in many complex matrix operations, including certain statistical applications. For instance, if $|\mathbf{A}| = 0$, then one or more columns of $\mathbf{A}$ can be expressed as a linear combination of other columns; that is, the columns are linearly dependent. As a simple example, with

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & 4 \\ 4 & 1 & 5 \\ 2 & 3 & 5 \end{bmatrix},$$

it is easy to see that the third column equals the sum of the first two, and indeed the determinant of this matrix is 0. Other examples are less obvious, of course. In general, a matrix with a determinant equal to 0 is called a **singular** matrix, implying that there is a linear dependency among its columns. Later, in Chapters 8 and 9, determinants of covariance matrices will appear in the maximum likelihood fitting function for factor analysis and structural equation modeling.

For a (2×2) matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

the determinant is calculated as $|\mathbf{A}| = (a_{11}a_{22}) - (a_{12}a_{21})$. Next, for a (3×3) matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

the formula is

$$|\mathbf{A}| = (a_{11}a_{22}a_{33}) - (a_{11}a_{23}a_{32}) + (a_{12}a_{23}a_{31}) - (a_{12}a_{21}a_{33}) + (a_{13}a_{21}a_{32}) - (a_{13}a_{22}a_{31}).$$

For the (3×3) matrix in the previous paragraph, applying this formula obtains

$$|\mathbf{A}| = (15) - (45) + (10) - (20) + (48) - (8) = 0,$$

confirming that the matrix is singular. For larger matrices, the formula is progressively more complex, such that hand calculation is excessively tedious.

Determinants are involved in the calculation of a matrix **inverse**. In matrix algebra, there is no direct analog of division, but many *square* matrices have an inverse, which then can be

used to solve equations, as shown below. In scalar arithmetic, division can also be expressed using inverses. For example, the expression

$$(35 \div 7) = (35 \times \frac{1}{7}) = 5$$

is equivalent to the expression

$$35(7^{-1}) = 5$$

where $\frac{1}{7} = 7^{-1}$ is the inverse of the scalar value 7. Note that, of course, the inverse of any scalar

value n has the property that $n^{-1}n = 1$; for example, $7 \times 7^{-1} = 1$. Analogously, the inverse of a square matrix $\mathbf{A}$ is itself a square matrix of the same dimension, written $\mathbf{A}^{-1}$, with the property that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$.

Finding the inverse of a square matrix is not simply a matter of taking the inverse of each individual scalar element within the matrix. The correct process is too mathematically complex for me to present here (see Carroll et al., 1997, pp. 136-143). But we can use matrix multiplication to establish whether one matrix is an inverse of another. For example, the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix}$$

has an inverse of

$$\mathbf{A}^{-1} = \begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix}$$

because

$$\mathbf{A}\mathbf{A}^{-1} = \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}$$

and

$$\mathbf{A}^{-1}\mathbf{A} = \begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}.$$

But it is important to recognize that not all square matrices have inverses. If a square matrix does have a unique inverse, then the matrix is termed **nonsingular**; a square matrix *without* an inverse (corresponding to having a determinant equal to zero) is called singular. Singular matrices can causes problems in certain statistical applications, in which case software applications typically

produce a warning message that a particular matrix related to the model being estimated is singular.

The ability to multiply matrices together and to take inverses of square matrices makes it simple to formulate *systems* of scalar equations, such as a set of several regression equations, as a *single* matrix equation, and then to solve for the unknown quantities in the system. For a brief example, consider the following system of two scalar linear equations with two unknowns, x_1 and x_2 :

$$-5 = 2x_1 + 9x_2$$

$$7 = x_1 + 4x_2 = 7$$

Next, these *two* scalar equations are expressed as a *single* matrix equation:

$$\begin{bmatrix} -5 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

or, more compactly,

$$\mathbf{y} = \mathbf{A}\mathbf{x}.$$

To solve for $\mathbf{x}$, we can pre-multiply both sides of the matrix equation by $\mathbf{A}^{-1}$:

$$\mathbf{A}^{-1}\mathbf{y} = \mathbf{A}^{-1}\mathbf{A}\mathbf{x},$$

which implies

$$\mathbf{A}^{-1}\mathbf{y} = \mathbf{I}\mathbf{x}$$

or simply

$$\mathbf{A}^{-1}\mathbf{y} = \mathbf{x}.$$

Thus, the solution for $\mathbf{x}$ is

$$\begin{bmatrix} -4 & 9 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} -5 \\ 7 \end{bmatrix} = \begin{bmatrix} 83 \\ -19 \end{bmatrix} = \mathbf{x}.$$

Matrix Calculations for Statistical Applications

Means

Recall that the formula for calculating a sample mean using scalar terms is

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i.$$

For instance, take a very small sample of $N = 5$ with X_i scores of $[3, 4, 6, 2, 5]$. The mean of these scores is thus

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i = \frac{1}{5} (3 + 4 + 6 + 2 + 5) = \frac{1}{5} (20) = 4.$$

To get the sample mean of a single variable X_i using matrix algebra, first express the data (i.e., the sample values of X_i) as a column vector:

$$\mathbf{X} = \begin{bmatrix} 3 \\ 4 \\ 6 \\ 2 \\ 5 \end{bmatrix}.$$

To add up all the elements in a column vector $\mathbf{X}$, we need to pre-multiply $\mathbf{X}$ by the conformable **unit vector**, $\mathbf{1}'$. A unit vector is simply a vector of ones, which in the current example must have a length of 5 to be conformable with our data vector $\mathbf{X}$:

$$\mathbf{1}' = [1 \quad 1 \quad 1 \quad 1 \quad 1].$$

Then it is relatively easy to see that the matrix expression for the summation of the data is

$$\sum_{i=1}^N X_i = \underset{(1 \times N)}{\mathbf{1}'} \underset{(N \times 1)}{\mathbf{X}}.$$

Thus, the matrix formula for the sample mean of a single variable, X , is

$$\bar{X} = \frac{1}{N} \mathbf{1}' \mathbf{X}.$$

With the example data, the matrix calculation of the mean is

$$\bar{X} = \frac{1}{5} [1 \quad 1 \quad 1 \quad 1 \quad 1] \begin{bmatrix} 3 \\ 4 \\ 6 \\ 2 \\ 5 \end{bmatrix} = \frac{1}{5} [20] = 4.$$

Next, rather than a single variable, consider a $(N \times P)$ data matrix $\mathbf{X}$, containing data for P variables. For example, $\mathbf{X}$ could contain scores for $N = 5$ participants on $P = 3$ variables, X_1 , X_2 , and X_3 :

$$\mathbf{X} = \begin{bmatrix} 3 & 7.8 & 0 \\ 4 & 8.0 & 0 \\ 6 & 8.2 & 1 \\ 2 & 7.9 & 1 \\ 5 & 8.1 & 1 \end{bmatrix}.$$

We would like to calculate a mean for each column (i.e., each variable) of $\mathbf{X}$. These three means will be collected in the vector $\bar{\mathbf{X}}' = [\bar{X}_1 \quad \bar{X}_2 \quad \bar{X}_3]$. The matrix formula for calculating a vector of means from a $(N \times P)$ data matrix is

$$\bar{\mathbf{X}}' = \frac{1}{N} \underset{(1 \times P)}{\mathbf{1}'} \underset{(1 \times N)}{\mathbf{X}} \underset{(N \times P)}{.}$$

Notice that this formula is a simple generalization of that given above for calculating the mean of a single variable, in that the dimension of the data matrix has expanded from $(N \times 1)$ to $(N \times P)$.

With the example data matrix, the means are calculated as

$$\bar{\mathbf{X}}' = \frac{1}{5} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 7.8 & 0 \\ 4 & 8.0 & 0 \\ 6 & 8.2 & 1 \\ 2 & 7.9 & 1 \\ 5 & 8.1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 8.0 & 0.6 \end{bmatrix} = [\bar{X}_1 \quad \bar{X}_2 \quad \bar{X}_3].$$

Variance and covariance

Using scalar terms, the formula for the unbiased sample variance of a single variable X_i is

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2.$$

An analogous matrix algebra approach to calculating the variance of X involves first calculating a vector of *deviation scores*, $\mathbf{X}_D$:

$$\mathbf{X}_D = \mathbf{X} - \mathbf{1}\bar{X}.$$

For example, using the same data vector given earlier,

$$\mathbf{X} = \begin{bmatrix} 3 \\ 4 \\ 6 \\ 2 \\ 5 \end{bmatrix}, \text{ which has } \bar{X} = 4,$$

the deviation scores are

$$\mathbf{X}_D = \begin{bmatrix} 3 \\ 4 \\ 6 \\ 2 \\ 5 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \\ 2 \\ 5 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ -2 \\ 1 \end{bmatrix}.$$

Given the deviation score vector, the matrix formula for the unbiased sample variance of X is

$$s^2 = \frac{1}{N-1} \mathbf{X}'_D \mathbf{X}_D.$$

So with the example data, we have

$$s^2 = \frac{1}{4} \begin{bmatrix} -1 & 0 & 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 2 \\ -2 \\ 1 \end{bmatrix} = \frac{1}{4} 10 = 2.5.$$

Next, again consider a complete $(N \times P)$ data matrix $\mathbf{X}$. To get the deviation scores for each column (i.e., each variable) in $\mathbf{X}$, generalize the deviation score formula from the previous paragraph to apply to a data matrix of dimension $(N \times P)$ rather than just $(N \times 1)$:

$$\mathbf{X}_D = \underset{(N \times P)}{\mathbf{X}} - \underset{(N \times P)}{\mathbf{1}} \underset{(N \times 1)(1 \times P)}{\bar{\mathbf{X}}'}.$$

Using the same data matrix given earlier,

$$\mathbf{X} = \begin{bmatrix} 3 & 7.8 & 0 \\ 4 & 8.0 & 0 \\ 6 & 8.2 & 1 \\ 2 & 7.9 & 1 \\ 5 & 8.1 & 1 \end{bmatrix}, \text{ which has } \bar{\mathbf{X}}' = [4 \quad 8.0 \quad 0.6],$$

the matrix of deviation scores is

$$\mathbf{X}_D = \begin{bmatrix} 3 & 7.8 & 0 \\ 4 & 8.0 & 0 \\ 6 & 8.2 & 1 \\ 2 & 7.9 & 1 \\ 5 & 8.1 & 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} [4 \quad 8.0 \quad 0.6],$$

which is

$$\mathbf{X}_D = \begin{bmatrix} 3 & 7.8 & 0 \\ 4 & 8.0 & 0 \\ 6 & 8.2 & 1 \\ 2 & 7.9 & 1 \\ 5 & 8.1 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 8 & .6 \\ 4 & 8 & .6 \\ 4 & 8 & .6 \\ 4 & 8 & .6 \\ 4 & 8 & .6 \end{bmatrix} = \begin{bmatrix} -1 & -0.2 & -0.6 \\ 0 & 0.0 & -0.6 \\ 2 & 0.2 & 0.4 \\ -2 & -0.1 & 0.4 \\ 1 & 0.1 & 0.4 \end{bmatrix}.$$

Now, given the matrix of deviation scores, the matrix formula

$$\mathbf{S} = \frac{1}{(N-1)} \mathbf{X}_D' \mathbf{X}_D$$

produces the complete **variance-covariance matrix** for data matrix $\mathbf{X}$. Because matrix $\mathbf{X}_D$ has dimension $(N \times P)$, it follows that the product $\mathbf{X}_D' \mathbf{X}_D$ will have dimension $(P \times P)$, and thus the variance-covariance matrix $\mathbf{S}$ will also have dimension $(P \times P)$. Applying the example $\mathbf{X}_D$ from the previous paragraph,

$$\mathbf{S} = \frac{1}{N-1} \mathbf{X}_D' \mathbf{X}_D = \frac{1}{4} \begin{bmatrix} -1.0 & 0.0 & 2.0 & -2.0 & 1.0 \\ -0.2 & 0.0 & 0.2 & -0.1 & 0.1 \\ -0.6 & -0.6 & 0.4 & 0.4 & 0.4 \end{bmatrix} \begin{bmatrix} -1 & -0.2 & -0.6 \\ 0 & 0.0 & -0.6 \\ 2 & 0.2 & 0.4 \\ -2 & -0.1 & 0.4 \\ 1 & 0.1 & 0.4 \end{bmatrix},$$

which produces

$$\mathbf{S} = \frac{1}{4} \begin{bmatrix} 10.0 & 0.9 & 1.0 \\ 0.9 & 0.1 & 0.2 \\ 1.0 & 0.2 & 1.2 \end{bmatrix} = \begin{bmatrix} 2.5 & 0.225 & 0.25 \\ 0.225 & 0.025 & 0.05 \\ 0.25 & 0.05 & 0.3 \end{bmatrix}.$$

In a variance-covariance matrix $\mathbf{S}$, the diagonal elements are the variances of each column (i.e., each variable) in the original data matrix $\mathbf{X}$. For example, element s_{11} is the variance of X_1 , where X_1 is the variable represented in the first column of $\mathbf{X}$. The off-diagonal elements are *product-moment covariances*, each of which summarizes the linear relationship between two variables, as described in Chapter 1. For example, element s_{21} is the covariance

between X_2 and X_1 . A variance-covariance matrix is usually simply called a *covariance matrix*. Thus, it is important to keep in mind that a covariance matrix still has variances as its diagonal elements. In this sense, a variance can be thought of as “the covariance of a variable with itself.”

Standard deviation

Expressed using scalar terms, standard deviation s is simply the square-root of variance:

$$s = \sqrt{s^2} = (s^2)^{1/2}.$$

To get the standard deviations of each column (i.e., each variable) in a data matrix $\mathbf{X}$, we first need to define a diagonal matrix $\mathbf{D}$ which contains only the diagonal elements of the corresponding covariance matrix $\mathbf{S}$ calculated from $\mathbf{X}$:

$$\mathbf{D} = \text{diag}(\mathbf{S}).$$

Following from the example covariance matrix above,

$$\mathbf{D} = \text{diag}(\mathbf{S}) = \text{diag} \begin{bmatrix} 2.5 & 0.225 & 0.25 \\ 0.225 & 0.025 & 0.05 \\ 0.25 & 0.05 & 0.3 \end{bmatrix} = \begin{bmatrix} 2.5 & 0 & 0 \\ 0 & 0.025 & 0 \\ 0 & 0 & 0.3 \end{bmatrix}.$$

The standard deviations of the variables in $\mathbf{X}$ are then the diagonal elements of the matrix $\mathbf{D}^{1/2}$:

$$\mathbf{D}^{1/2} = \begin{bmatrix} 2.5^{1/2} & 0 & 0 \\ 0 & 0.025^{1/2} & 0 \\ 0 & 0 & 0.3^{1/2} \end{bmatrix} = \begin{bmatrix} 1.58 & 0 & 0 \\ 0 & 0.16 & 0 \\ 0 & 0 & 0.55 \end{bmatrix}$$

(with rounding to two decimal places).

Correlations

As presented in Chapter 1 (Equation 1.9), the sample product-moment correlation between two variables, X and Y , can be calculated using scalar terms with

$$r_{XY} = \frac{s_{XY}}{\sqrt{s_X^2} \sqrt{s_Y^2}} = \frac{s_{XY}}{s_X s_Y},$$

where s_X^2 and s_Y^2 are the sample variances of X and Y , s_X and s_Y are the standard deviations, and s_{XY} is the product-moment covariance between X and Y . To see how this scalar formula generalizes to a matrix formula, it is helpful to rearrange its terms as follows:

$$r_{XY} = \frac{s_{XY}}{\sqrt{s_X^2} \sqrt{s_Y^2}} = s_{XY} (s_X^2)^{-1/2} (s_Y^2)^{-1/2},$$

which then can be written as

$$r_{XY} = (s_X^2)^{-1/2} s_{XY} (s_Y^2)^{-1/2}.$$

The formula for converting a $P \times P$ sample covariance matrix, $\mathbf{S}$, for P variables into the *correlation matrix*, $\mathbf{R}$, for these variables has an analogous form:

$$\mathbf{R} = \mathbf{D}^{-1/2}(\mathbf{S}) \mathbf{D}^{-1/2},$$

where $\mathbf{D}$ is the $P \times P$ diagonal matrix of variances as defined above. By this process, the diagonal elements of $\mathbf{R}$ will always equal 1 and the off-diagonal elements will always fall between -1 and +1.

Following from the same three-variable running example used in the previous sections, we have

$$\mathbf{D}^{-1/2} = \begin{bmatrix} \frac{1}{2.5^{1/2}} & 0 & 0 \\ 0 & \frac{1}{0.025^{1/2}} & 0 \\ 0 & 0 & \frac{1}{0.3^{1/2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{1.58} & 0 & 0 \\ 0 & \frac{1}{0.16} & 0 \\ 0 & 0 & \frac{1}{0.55} \end{bmatrix},$$

which leads to

$$\mathbf{R} = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2} = \begin{bmatrix} \frac{1}{1.58} & 0 & 0 \\ 0 & \frac{1}{0.16} & 0 \\ 0 & 0 & \frac{1}{0.55} \end{bmatrix} \begin{bmatrix} 2.5 & 0.225 & 0.25 \\ 0.225 & 0.025 & 0.05 \\ 0.25 & 0.05 & 0.3 \end{bmatrix} \begin{bmatrix} \frac{1}{1.58} & 0 & 0 \\ 0 & \frac{1}{0.16} & 0 \\ 0 & 0 & \frac{1}{0.55} \end{bmatrix}.$$

Thus,

$$\mathbf{R} = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2} = \begin{bmatrix} 1.00 & .90 & .29 \\ .90 & 1.00 & .58 \\ .29 & .58 & 1.00 \end{bmatrix},$$

which is indeed a symmetric matrix with diagonal elements equal to 1 and off-diagonal elements being product-moment correlations between -1 and +1.

Matrix equations for linear regression

The multiple linear regression model with one outcome variable and P regressors can be expressed in matrix form as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\mathbf{Y}$ is a vector of outcome variable scores for N cases or individuals, $\mathbf{X}$ is a $(N \times (P + 1))$ matrix of scores on the P regressors (as well as a column containing a constant equal to 1), $\boldsymbol{\beta}$ is a vector of $(P + 1)$ population linear slope coefficients (including an intercept parameter), and $\boldsymbol{\varepsilon}$ is a vector of error terms for the N cases. Written another way, this matrix equation (for $P > 2$) is

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1P} \\ 1 & x_{21} & x_{22} & \cdots & x_{2P} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{N1} & x_{N2} & \cdots & x_{NP} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_P \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{bmatrix}.$$

It is critical to realize that the matrix of regressor values, $\mathbf{X}$ (which is often referred to as the **design matrix**) has as its first column a constant value equal to 1 for all rows (i.e., all N cases).

This constant term is included so that the intercept parameter, β_0 , is used to model $\hat{Y}$ for each case. One way to think about the intercept is that it is the regression coefficient for a “variable” which is actually a constant value ($= 1$) for each individual case. To illustrate, recall that a matrix equation is essentially a collection of separate scalar equations. Here, the regression matrix equation is a collection of N individual linear equations, one for each of $i = 1, 2, \dots, N$ individuals. For a generic individual case i , the regression equation is

$$y_i = \beta_0 1 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_P x_{iP} + \varepsilon_i,$$

which, of course, is mathematically the same as the more familiar expression

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_P x_{iP} + \varepsilon_i.$$

The only difference between these two scalar equations is that the first explicitly indicates that the regression intercept parameter is the coefficient for a “variable” which is set to equal 1 for all N cases. Therefore this “variable” is actually a *constant*, not a variable, which is why output from statistical software often reports the intercept estimate on a row labeled “Constant” rather than “Intercept.”

Recall from Chapter 2 that the regression parameters, that is, the intercept and slope coefficients in $\boldsymbol{\beta}$, are usually estimated from sample data using the ordinary least squares (OLS)

method, which minimizes the sum of the squared residuals, SS_e . In matrix terms, the OLS solution for β is obtained by minimizing the result of the (estimated) error vector $\hat{\epsilon}$ pre-multiplied by its transpose; it turns out that $\hat{\epsilon}'\hat{\epsilon} = SS_e$. Because the estimated regression model

$$Y = X\hat{\beta} + \hat{\epsilon}$$

implies

$$\hat{\epsilon} = Y - X\hat{\beta},$$

minimizing $\hat{\epsilon}'\hat{\epsilon}$ is equivalent to finding a vector of parameter estimates, $\hat{\beta}$, which minimizes the value of f in the matrix function

$$f = (Y - X\hat{\beta})'(Y - X\hat{\beta}).$$

Solving this minimization problem (which requires calculus) leads to the OLS solution

$$\hat{\beta} = (X'X)^{-1}X'Y.$$

The $(N \times 1)$ vector of outcome variable predicted values, $\hat{Y}$, from the estimated regression model is thus

$$\hat{Y} = X\hat{\beta} = X(X'X)^{-1}X'Y,$$

or simply $\hat{Y} = HY$ where $H = X(X'X)^{-1}X'$. This matrix H is commonly called the **hat matrix**.

Another name for the leverage of a given case is *hat value*; the reason for this term is that the leverage values are the diagonal elements of the hat matrix.

Eigenvalues and eigenvectors

Many multivariate statistical procedures rely on calculation of the **eigenstructure** of certain symmetric matrices, such as covariance and correlation matrices. The eigenstructure of a symmetric matrix S can be expressed as

$$(1) \quad S = UDU',$$

where the columns of U are **eigenvectors** and D is a diagonal matrix with **eigenvalues** as diagonal elements (this D is not the same as the D matrix used earlier to calculate a covariance matrix). Eigenvalues and eigenvectors are also sometimes called *characteristic roots and vectors*.

Taking a few steps back, if S is a $(P \times P)$ symmetric matrix, u is a vector of length P , and λ is a scalar such that

$$(2) \quad \mathbf{S}\mathbf{u} = \lambda\mathbf{u},$$

then λ is an eigenvalue of $\mathbf{S}$ and $\mathbf{u}$ is the corresponding eigenvector. The potential value of this relationship may seem readily apparent: On the left-hand side of the equation, $\mathbf{u}$ is pre-multiplied by a full $(P \times P)$ matrix (i.e., $\mathbf{S}$), but on the right-hand side, this matrix has been replaced by a single scalar term (i.e., λ). It is as if all of the numerical information in $\mathbf{S}$ has been collapsed into a single value, λ .

But for any given symmetric matrix, there will generally be more than one non-zero eigenvalue-eigenvector combination which satisfies Equation 2. Specifically, a $(P \times P)$ symmetric matrix has P eigenvalues which are typically ordered from largest to smallest, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_P$, and are respectively associated with eigenvectors

$\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P$.

The eigenvalues are arranged in descending order as the diagonal elements of a diagonal matrix

$$\mathbf{D} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_P \end{bmatrix}$$

and the corresponding eigenvectors can be collected as columns in a matrix $\mathbf{U}$ (such that the first column of $\mathbf{U}$ is $\mathbf{u}_1$, the second column is $\mathbf{u}_2$, and so on). Next, Equation 2 can be expanded to $\mathbf{SU} = \mathbf{UD}$,

now showing that the original symmetric matrix multiplied by the *set* of eigenvectors can be expressed as the product of the eigenvectors and a simple diagonal matrix containing the eigenvalues. The matrix of eigenvectors $\mathbf{U}$ is *orthogonal*, meaning that $\mathbf{UU}' = \mathbf{I}$. Consequently, the latter equation can be easily rearranged to produce Equation 1, the eigenstructure equation for symmetric matrix $\mathbf{S}$.

If all eigenvalues of a symmetric matrix are positive, then the matrix is considered **positive definite**. Certain statistical estimation algorithms require positive-definite covariance (or correlation) matrices, because non-positive definite matrices are singular and cannot be inverted. It happens quite often that statistical software procedures produce error messages indicating that results could not be obtained (or that the results that *were* obtained may be incorrect) because a particular matrix is non-positive definite or singular.

References

Carroll, J.D., Greene, P.E., & Chaturvedi, A. (1997). *Mathematical tools for applied multivariate analysis* (revised edition). Academic Press.

Fox, J. (2009). *A mathematical primer for social statistics*. Sage Publications.